

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester V Examination, 2013-14

Paper No. GLB-501: Physical and Structural Geology

Time: 3 Hours

Maximum Marks: 70

Ref. No.: BS/Sem V/257

*Instructions: Attempt any **five** questions in all, at least **two** questions each from Section A and B. Section C is compulsory. All questions carry equal marks (14 marks each).*

SECTION — A

Question 1. What is a fold? Give the genetic classification of folds and discuss the mechanism of their development. [14]

Question 2. Describe the erosional and depositional features formed by the wind. [14]

Question 3. Discuss the following: [2 × 7 = 14]

- (a) Continental drift theory of Taylor
- (b) Effect of faulting on folded outcrop

SECTION — B

Question 4. Write short notes on the following: [4 × 3.5 = 14]

- (a) Airy's and Pratt's hypothesis of Isostasy
- (b) Erosional geomorphic features of river in young stage (upper course)
- (c) Landscape Plain
- (d) Flood and their mitigation

Question 5. Discuss with suitable examples: [2 × 7 = 14]

- (a) Depositional geomorphic features formed by glacier
- (b) Briefly describe the different types of earth movements

Question 6. Discuss the following: [2 × 7 = 14]

- (a) Types of unconformities and their geological significance
- (b) Criteria of recognition of faults in the field and on the maps

Question 7. Differentiate between the following: [4 × 3.5 = 14]

- (a) Slaty cleavage and strain slip cleavage
- (b) Dip fault and dip slip fault
- (c) Klippe and Window
- (d) Isoclinal and Monoclinical fold

SECTION — C (Compulsory)

Question 8. Describe each of the following in two to three sentences. Give suitable diagrams where applicable: [7 × 2 = 14]

- (i) Difference between active remote sensing and passive remote sensing
- (ii) Distinction within Bouguer anomaly and topographic anomaly
- (iii) Define the Median Mass
- (iv) Listric fault
- (v) Horst and Graben
- (vi) Autochthonous and allochthonous beds
- (vii) Positive and negative flower structure